

Activity Design

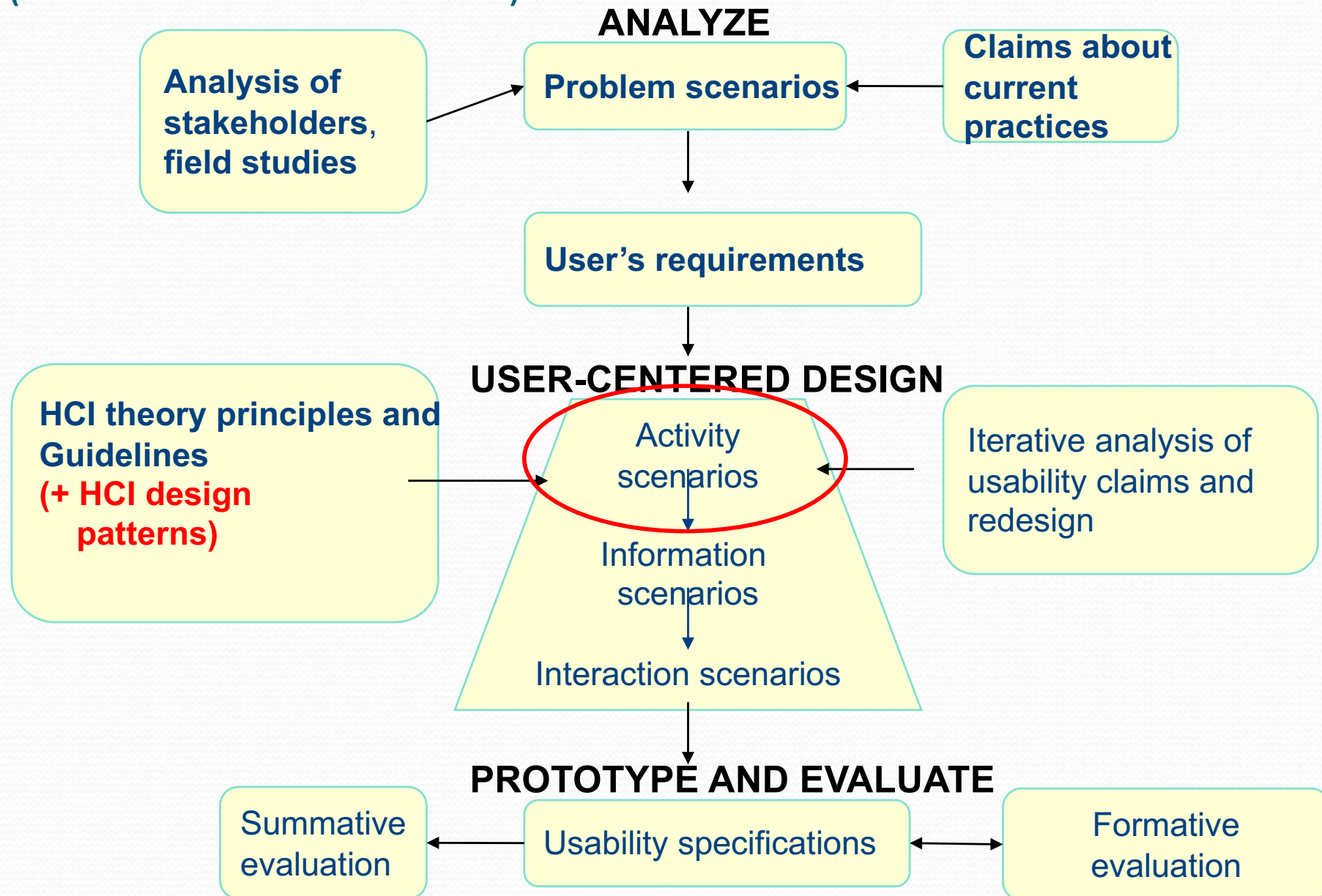
Corso di Informatica – Laurea Magistrale

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A scenario-based design process

(Rosson and Carroll 2002)



From Requirements to Design

Goal: work from problems and opportunities of problem domain to envision new activities

- requirements analysis sets the scene
- design transforms people's activities
 - new technology, new tasks, new experiences
 - and the cycle continues...

Problem claims:
look for design
ideas that address
negatives, but
keep positives

Problem scenarios: work from
current practice to build new

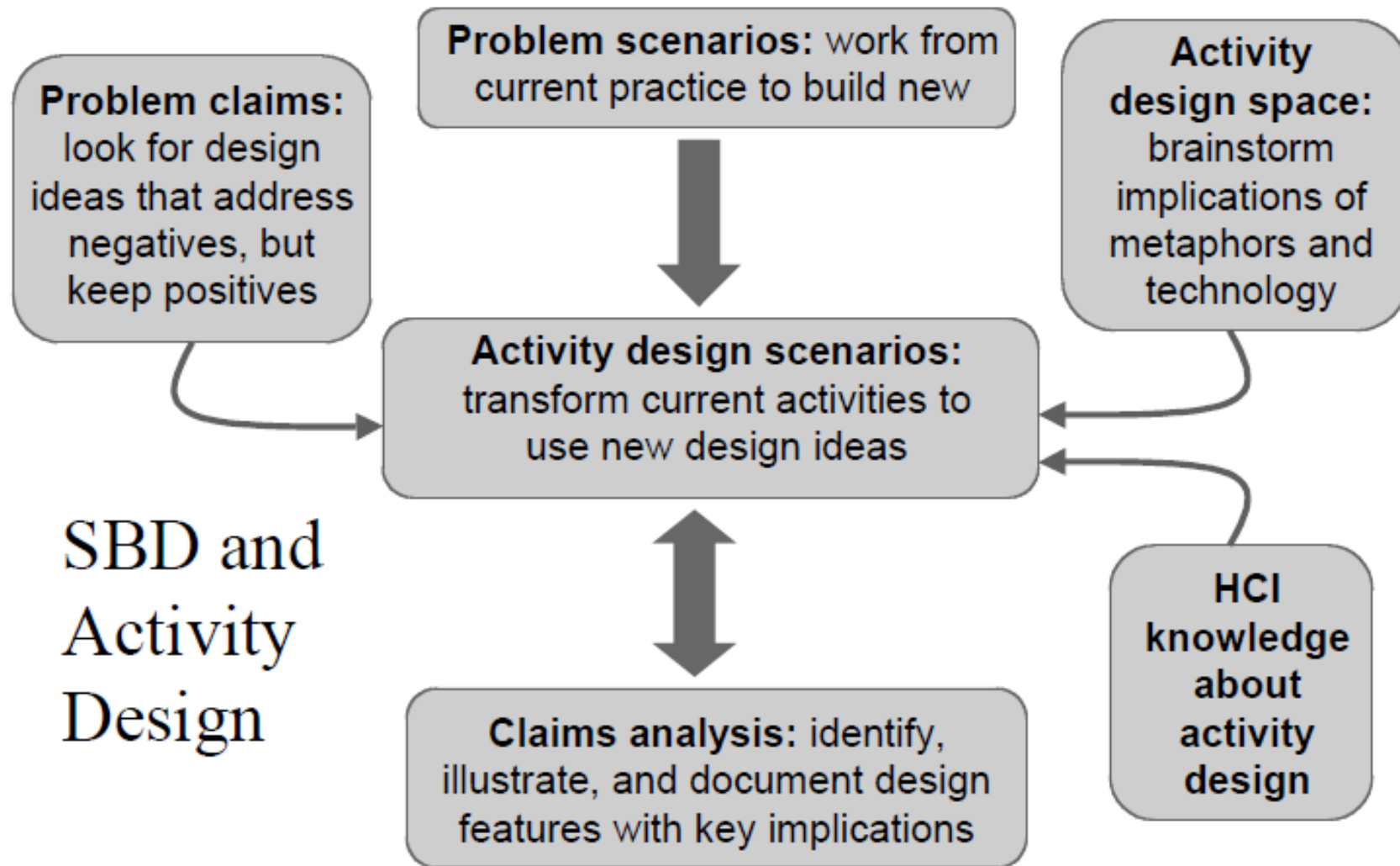
**Activity
design space:**
brainstorm
implications of
metaphors and
technology

Activity design scenarios:
transform current activities to
use new design ideas

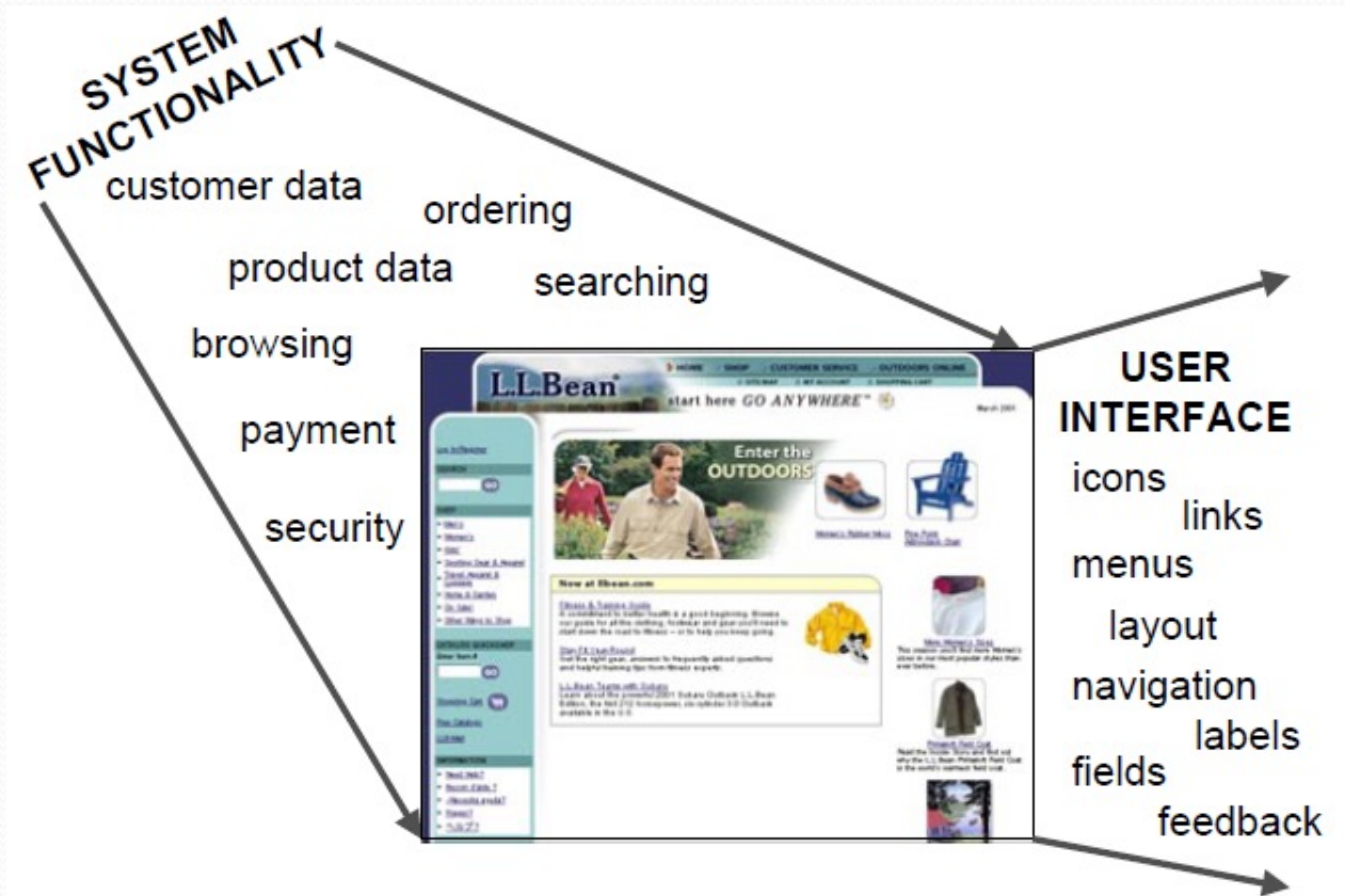
**HCI
knowledge
about
activity
design**

Claims analysis: identify,
illustrate, and document design
features with key implications

SBD and
Activity
Design



The Two Faces of HCI Design





Why system functionality first?

- It determines what is possible
- We can focus on *what* a system will do and not worry about how
- Considering system functionality first, designers can make progress more quickly
- User interface design can better proceed if it is clear what the system will do and why.
- Activity design is also called “conceptual design” or “task-level design”.



Envisioning New Activities

Three activity design concerns to keep in mind:

- Effectiveness: Designing tasks that meet real needs
- Comprehension: Designing concepts and services that your users can predict, understand
- Satisfaction: Designing tasks that are motivating and lead to feelings of accomplishment, satisfaction



Designing for Effectiveness

- Innovation is good, but how much is too much?
 - build on what is already working well
 - engage stakeholders in *cooperative (participatory) design*
- What parts of a task to support via technology?
 - leverage other aspects of the work context, both people and things (*distributed cognition*)
- Balance tendency toward general solutions with the needs of specific tasks
 - predict and support exceptions, provide special cases for common or critical tasks

Designing for Comprehension

- Cannot directly observe comprehension
 - must rely on users' behaviors, reactions, comments
 - make inferences about their *mental models*
- Metaphors play a crucial role in this
 - designers explore metaphors to get new ideas
 - users evoke metaphors to understand new concepts
- Try to leverage users' existing knowledge
 - anticipate and support analogical reasoning but look for ways to “break” current understandings

Metaphors Brainstorming

In activity design metaphors help think outside the box

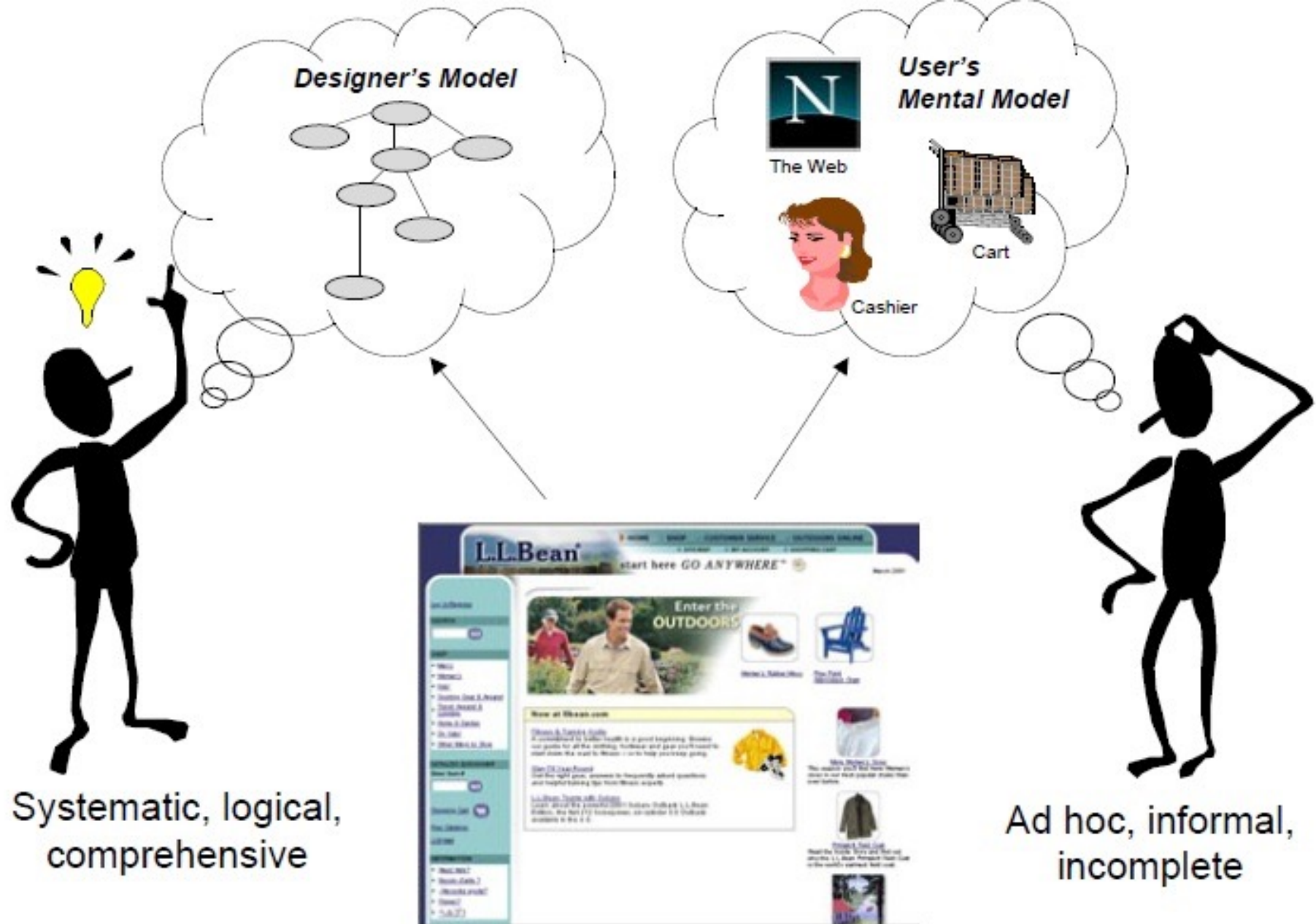
- Education, medicine, law, politics, finance, government, science, art
- Communities (physical, online)
- Teams, workgroups, task force
- Family, home, leisure
- Social relationships
- Religious or cultural value systems
- Crisis management, police
- And so on!!!

Ex: What if attending a virtual group meeting was like cleaning up a playground? Like watching a football game? Going out on a date?

Technology Brainstorming

- The world wide web – links, associations, navigation (“scent”)
- Intelligent systems – prediction, mixed initiative, automating, learning in place
- Programming – logic, decomposition, control flow, repetition, recursion, error detection and handling
- Pattern matching – abstraction, naming, classification, overlap
- End user applications – word processing, formatting, spreadsheet/finance, email, stats, scientific models, social networking, browsers

*Ex: Could a hypermedia database support interactions of a virtual team?
Can we imagine a role for an intelligent software agent? Are there online behavior patterns we might extract and present?*



Designing for Satisfaction

- Automate tedious tasks, but try not to remove sources of reward or accomplishment
 - carefully examine sources of reward, maintain or enhance opportunities for feelings of achievement
 - use the computer to make tasks more personal, more stimulating, more “fun”
- Balance the needs of individuals with those of the groups they work with
 - the people who do the most “work” when using a system may not be those who get the most “benefit”

Apply a Computational Metaphor

- Taking a software object's point of view
 - use *anthropomorphism to extend the task*
- Envision “scenario within a scenario”
 - ask: What would I do, if I were a shopping list trying to be helpful to this shopper?
- May discover new services, other objects
 - but need to reflect on implications of each idea
 - claims analysis helps you do this

HCI background – The active user

- Pursue goals that are meaningful given a personal, social and organizational context
 - Activity Theory, Situated Cognition
- Recruit and synthesize whatever goal-related resources we find in the environment or internally
 - Constructivistic learning, mental models & schemas, distributed cognition
- Problematize and decompose tasks when needed
 - Activities – action plans – operations
 - E.g. in response to “breakdowns” or “curiosity” episodes
- All these have paradoxical implications for design
 - Hard to get people to “pause & learn” what’s good for them
 - Prior knowledge helps but will interfere as well



Activity Scenarios

- The design team first introduces concrete ideas about new functionality, new ways of thinking about users' needs and how to meet them is the focus of activity scenarios.
- As in the other steps of the process, a **claims analysis** is generated to help identify the tradeoffs as you move forward with prototype



Working With Claims in Design

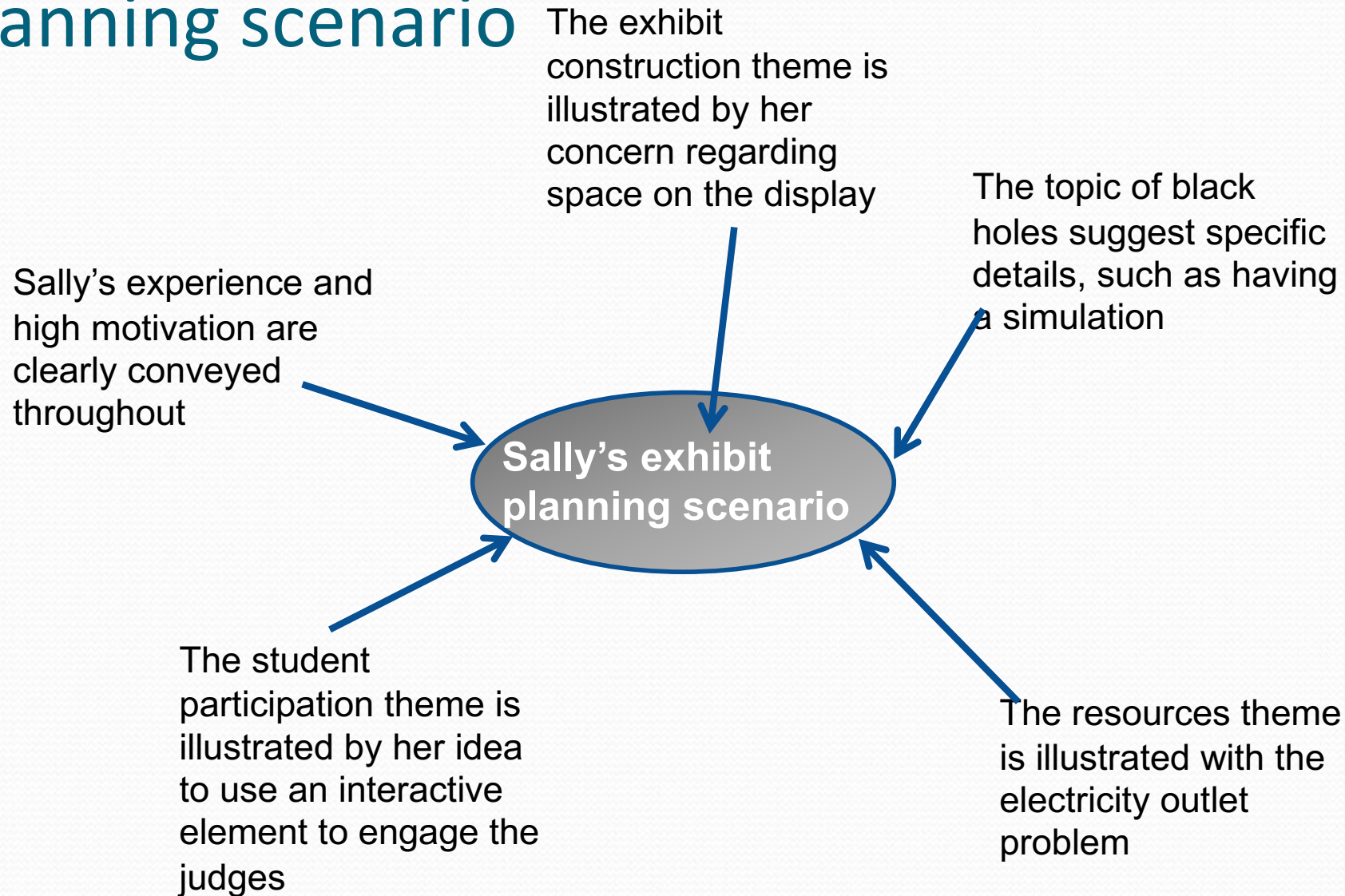
- Maintain or enhance as many positive impacts (upsides) as you can
- While removing or minimizing negative impacts (downsides) whenever possible
- As new ideas/scenarios are envisioned, consider also the new claims that they raise

Constructing a claim

- Reflect on key points in a scenario
 - Why was it written? Where are interesting things happening to actors? Why? Where does your starting “design lens” suggest?
- Extract what you believe may be causal features
 - In problem analysis, may NOT be technology mediated
 - In design, usually key concepts that you are exploring
- Hypothesize possible consequences of the features
 - Often leads you to go back and draw them out
 - Try to generate at least two contrasting implications
 - Use “What if?” thinking to help discover pros and cons
 - Consider through multiple theory lenses

```
<A feature of a situation or artifact in use...>  
+ <has positive effects for the actors>  
- <but may also have negative effects>
```

The Science Fair Case Study – ideas and concerns that contributed to the exhibit planning scenario



From the Science Fair Case Study

Giving all students the same physical display board

- + constrains and simplifies project lay-out
- + limits visual complexity for viewers
- but exhibits may have widely varying needs
- but regularity and simplicity may be boring to view

- How to maintain or enhance the benefits for planning and viewing but address the project-specific needs, or issues of variety and creativity in display?

The solution envisioned in one of the activity scenarios

*... When Sally goes to the exhibit construction area, **she finds a template with a suggested layout-title page, abstract, slide show, detailed results, project report, and bibliography.** At first she is worried that this will not fit the materials she has already created. But when she starts adding material, she can see that there is also still a lot of flexibility-for example, she can add a new component to hold her simulation. But she is not yet sure how she can share her physical star models...*

New Activities Lead to New Claims

We envisioned an exhibit template that would help in planning, but made it editable so that motivated (e.g. experienced) students could be as creative as desired

providing an exhibit template that is editable...

- + offers default planning and lay-out of exhibit parts
- + leverages students' familiarity with science projects
- + enables expert participants to be creative
- but students may trust the template too much
- but inexperienced students may become confused



Photo taken from a garden shop problem scenario: shopper wanders by, sees the pots, is reminded of needs, but store is out of the size pot she needs



From a Garden Shop Scenario

Susan ...She walks by the planter shelves, which remind her that she needs larger pots for some houseplants. A quick look reveals just one of the large size she needs, and it is too high for her to reach anyway ...

From a Garden Shop Scenario...

a wall of shelves for displaying products...

- + simplifies comparison of physical features
- + provides implicit availability/popularity indications
- + directs shoppers to related products
- but some shelves will be difficult to reach
- but browsing is constrained by physical layout

- How to maintain or enhance product comparisons and other analysis tasks, while addressing awkwardness and physical constraints?



The activity scenario

*... Susan decides to follow up on some recommended hand tools listed on the same page. She is taken immediately to some more information in what appears to be the **tools department** and decides to splurge on a new trowel...*

From a Garden Shop Scenario...

Organizing products by department.....

- + adheres to the store metaphor (people are used to shopping by department)
- + facilitates browsing when a user does not have a specific item in mind
- but user may not be able to match their target product with the department name
- but user may want to go directly to a particular target

Refining an Activity Design

- Ongoing claims analysis of activity scenarios
 - capture key ideas, begin to build *design rationale*
 - document problems to address during UI design
- Participatory design
 - brainstorming sessions with stakeholders
 - share rough ideas, get them to elaborate (metaphors can be very useful here as well)
- Consistency and coherence
 - reuse actors and objects to increase coherence
 - complement with ongoing “what if?” reasoning to expand and test the overall design